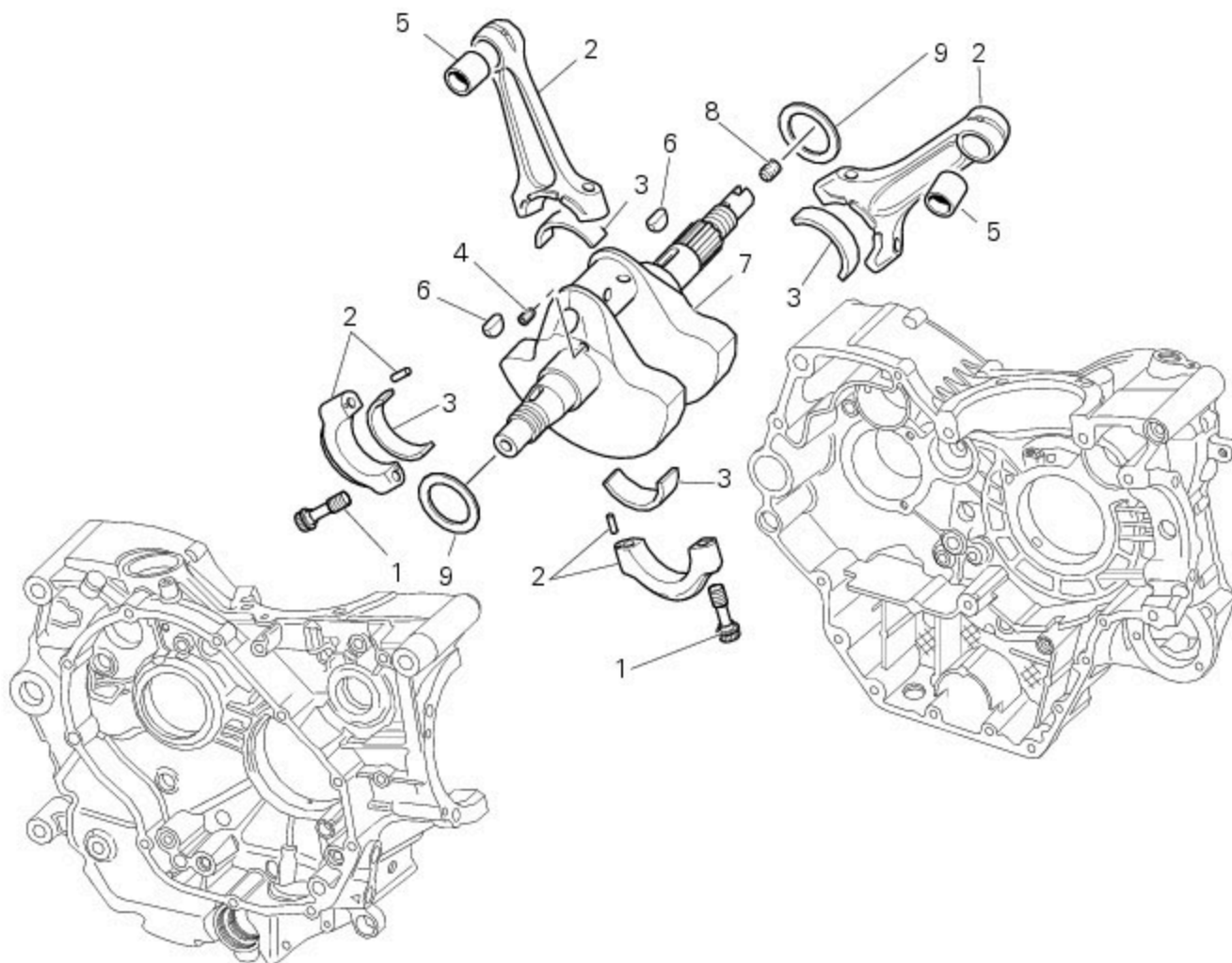


9.3 -Crankcase assembly: crankshaft/connecting rods assembly



- 1 Banjo bolt
- 2 Connecting rod assembly
- 3 Half bearing
- 4 Grub screw
- 5 Bushes
- 6 Key
- 7 Crankshaft
- 8 Grub screw
- 9 Shim washer

 Spare parts catalogue

1100 ABS [CONNECTING RODS](#)

1100 S ABS [CONNECTING RODS](#)



Important

Bold reference numbers in this section identify parts not shown in the figures alongside the text, but which can be found in the exploded view diagram.

Removal of the crankshaft/connecting rods assembly

Operation	Section reference
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Remove the engine from the frame	N 1, Removal of the engine
Remove the lubrication system	N 2.1, Removal of the oil pump
Remove the timing belts and the timing belt pulleys	N 4.2, Removal of the timing belt covers
Remove the camshafts	N 4.3, Removal of the camshafts
Remove the complete cylinder head unit	N 4.4, Removal of the cylinder head assembly
Remove the cylinder barrel/piston assemblies	N 5, Removal of the cylinder/piston assembly
Remove the alternator-side crankcase cover and the alternator assembly	N 8, Removal of the generator cover
Remove the mesh filter	D 4, Changing the engine oil and filter cartridge
Remove the complete clutch assembly	N 6.1, Removal of the clutch
Remove the clutch-side crankcase cover	N 6.2, Removal of the clutch cover
Remove the primary gear	N 6.3, Removal of the primary drive gear
Remove the starter motor cam chain idler gear	N 9.1, Removal of the starter motor idler gear
Remove the starter motor	P 3, Removal of the starter motor
Separate the crankcase halves	N 9.2, Separation of the crankcase halves

After separating the crankcase halves, withdraw the crankshaft (7) complete with connecting rods (2).

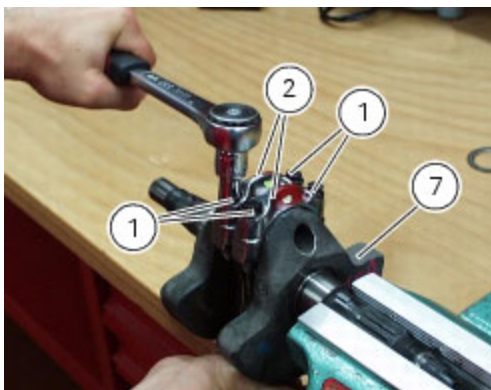
Disassembly of the crankshaft/connecting rods assembly

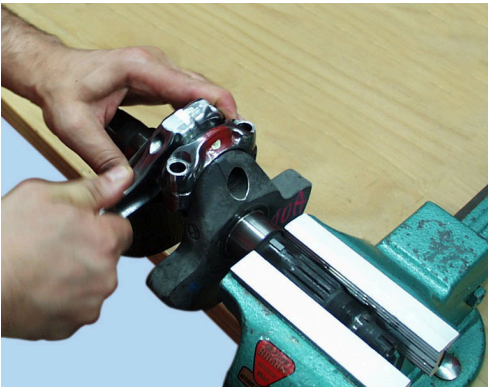
Unscrew the screws (1) and separate the connecting rods from the crankshaft.



Important

Take care not to mix up components of different connecting rods and maintain the original orientation.



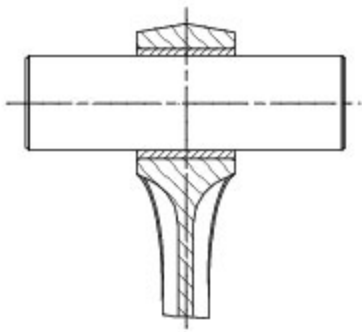


Overhaul of the connecting rods

Make the following dimensional checks on the connecting rods:

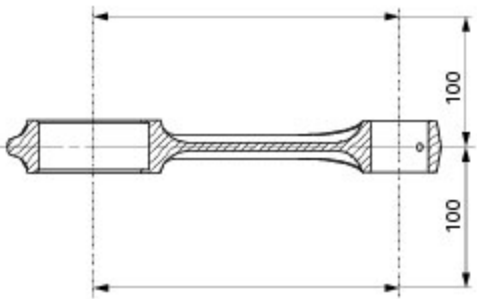
- clearance with gudgeon pin on assembly.

In the event of excessive wear (Sect. C 1.1 [Cylinder/Piston](#)), replace the connecting rod.



The small end bushing must be in good condition and firmly driven into its seat.

Check for parallelism error measured at **100** mm from the connecting rod longitudinal axis: the value must be **H-h** less than **0.02** mm; otherwise, renew the connecting rod.



The connecting rod is supplied in two size classes **A** and **B** relative to the big end diameter (Sect. C 1.1 [Cylinder/Piston](#)) as punch marked on the side of the cap.

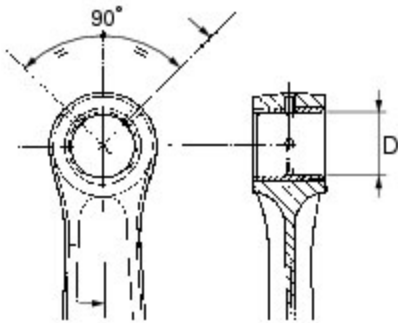
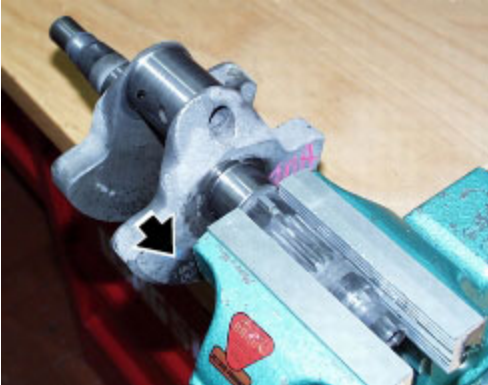
It is preferable to use crankshafts and connecting rods of the same size class.

Renewal of the small end bushing

To remove the worn bushing, use a suitable drift and a press.

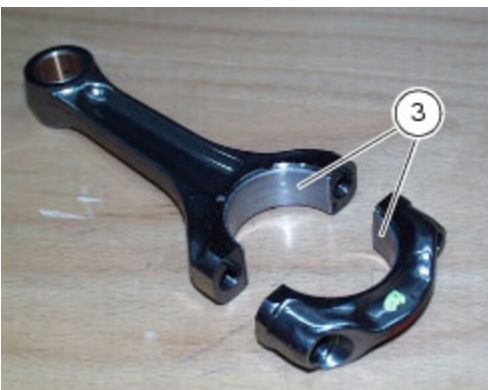
Drill lubrication holes into the new bushing in correspondence with the existing lubrication holes on the connecting rod small end.

Now ream out the bushing until the inside diameter (D) is **20.035** to **20.045** mm.

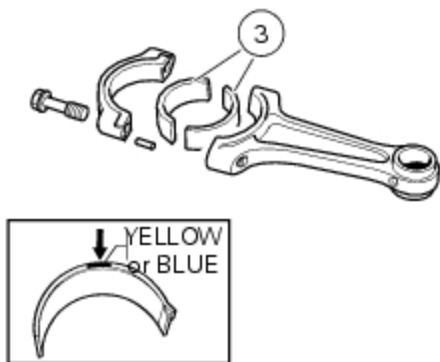


Renewal of the connecting rod bearings

It is good practice to renew the bearings (3) each time the engine is overhauled.



Replacement bearings are supplied ready for assembly and they must not be reworked with scrapers or emery cloth. The bearings may belong to two different size classes, each identified by a specific colour (YELLOW and BLUE).



The bearings are comprised of an external steel ring, the inner face of which is electro-plated with a lead-based compound.
The table shows the appropriate bearings to be fitted according to the size class of the crankshaft and connecting rod.

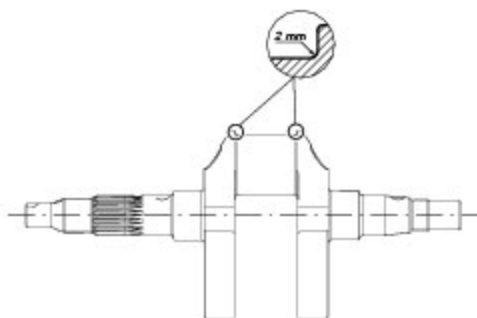
Crankshaft class	Connecting rod class	Bearings colour
A	A	BLUE + YELLOW
B	A	YELLOW + YELLOW
A	B	BLUE + BLUE
B	B	BLUE + YELLOW

Crankshaft overhaul

The main bearing and big-end journals should not be scored or grooved; the threads, keyways, and slots must be in good condition.

Check for fretting or burrs in the fillet between journal and shoulder.

Fillet radius: **2 mm**.

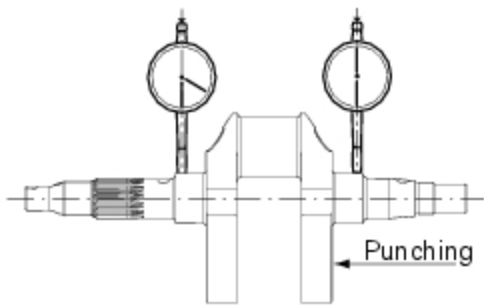


Use a micrometer to measure oval and taper of the crank pin. Measure oval and taper in several different directions.
Use a dial gauge to measure the alignment of the main journals by setting the crankshaft between two opposing centres.

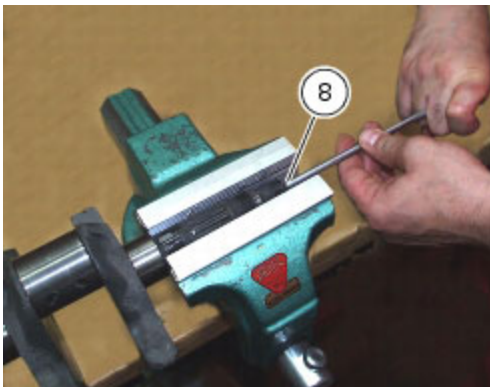
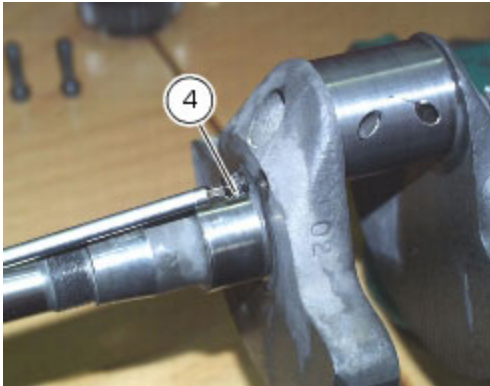
At each overhaul, it is advisable to clean the crankshaft's internal oilways.

The prescribed values are given in Sect. C 1.1, [Cylinder/Piston](#).

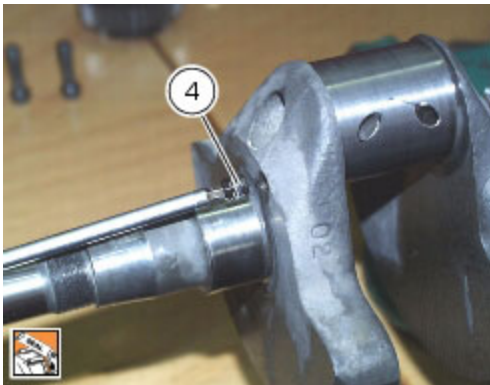
The crankshaft is supplied in two size classes (connecting rod pin) **A** and **B**, as punch marked on the side of the crank web on the pinion side.

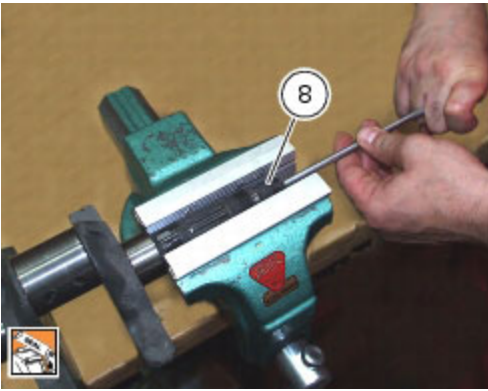


Unscrew all the plugs (4) and (8) from the crankshaft, heating the crankshaft, if necessary, to remove the threadlocker applied at the time of assembly.



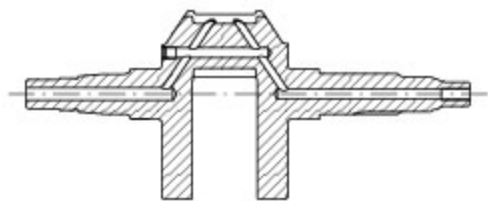
Clean all the oilways using suitable diameter metal brushes and then blow with compressed air to remove any residues that have accumulated and are restricting the oil flow.
Apply DUCAT liquid gasket to the threads of plugs (4) and (8) and refit them.





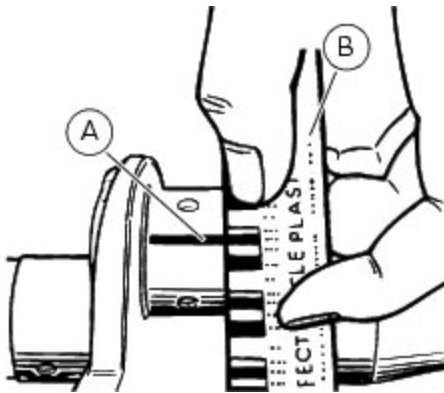
Lock the plugs to the specified torque (Sect. C 3, [Engine torque settings](#)).

Big-end bearing shell-journal clearance



To check the assembly clearance between the bearing shells and crankshaft journals you will lay a strip (A) of GREEN "Plastigage PG-1" on the journal. Fit the connecting rod with the original bearings and tighten the screws to a torque of **50 Nm**.

Remove the connecting rod and compare the thickness of the Plastigage strip to the scale (B).

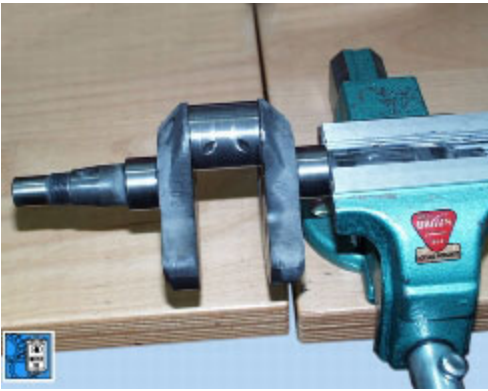


If the width measured corresponding to the existing clearance is not within the prescribed limit (Sect. C 1.1, [Cylinder/Piston](#)), either the bearings or the crankshaft must be replaced.

Reassembly of the connecting rods

Before starting, check that the crankshaft main bearing journals and big-end journals are free of burrs or evident signs of machining: if necessary, clean the surfaces with very fine emery cloth and oil.

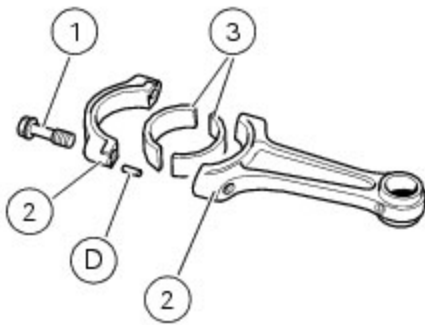
Check that the grooves are in perfect condition with no signs of forcing.



Check that each connecting rod (2) and cap are fitted with their locating pins (D).

Wash the pins and dry them with compressed air.

Clean and lubricate the journals and con-rod bearing shells (3) with engine oil and fit the connecting rods in their original mounting positions.



Fill the recommended grease into the two ends of the hole to lubricate threads and underside of the new screws (1) and the thread of the shaft.



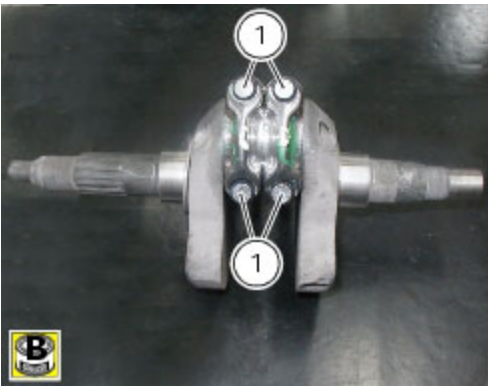
Warning

The grease utilised is an irritant in contact with the skin. Wear protective gloves.



Important

Lubrication of big-end cap screws is essential to obtain the correct coupling and to prevent breakage of the parts. The big-end cap bolts may only be used and tightened **once**, after which they must be discarded.



Interpose spacer part no. **88713.2878** between the connecting rods and eliminate any residual end float by inserting the fork feeler gauges coming with the tool: **0.1-0.2** and **0.3** mm thick.

Tighten the bolts (1) as specified:

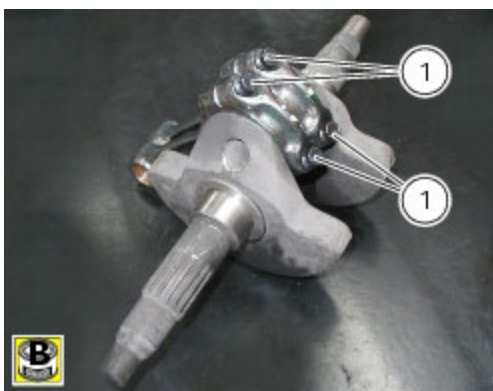
lubricate with the recommended grease;

tighten to **35** Nm, pause for 2 seconds and then back off through **360°**;

stage 1 tightening to a torque of **20** Nm;

stage 2 tightening to a torque of **35** Nm;

stage 3 tightening through **65°±0.5°** with torque check (**70** to **103** Nm).



Refitting the crankshaft/connecting rod assembly

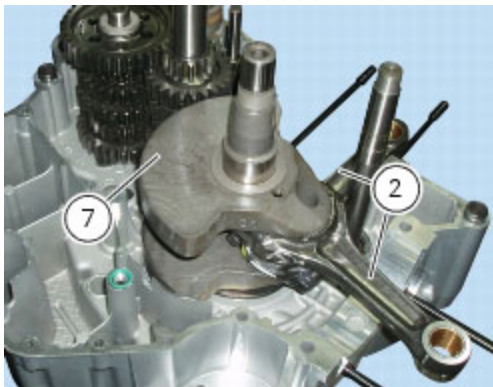
Install the connecting rod assembly (7) and (2) in the crankcase half.



Important

Make sure that the connecting rods (2) are correctly positioned in the cylinders. Incorrect positioning of the connecting rods at this stage will inevitably lead to the need to re-open the crankcase half.

Carry out the shimming procedure (Sect. N 9.2, [Shimming the shafts](#)).



Operation	Section reference
Reassemble the engine crankcase halves	N 9.2, Reassembly of the crankcase halves
Refit the starter motor	P 3, Refitting the starter motor
Refit the starter motor idler gear	N 9.1, Refitting the starter motor gear
Refit the primary gear	N 6.3, Fitting the primary drive gear pair and checking backlash
Refit the clutch-side crankcase cover	N 6.2, Refitting the clutch cover
Refit the clutch assembly	N 6.1, Reassembling the clutch
Refit the mesh filter	D 4, Changing the engine oil and filter cartridge
Refit the alternator-side crankcase cover and alternator assembly	N 8, Refitting the generator cover
Refit the cylinder barrel/piston assemblies	N 5, Refitting the cylinder/piston assembly
Refit the complete cylinder head unit	N 4.4, Refitting the cylinder head assembly
Refit the camshafts	N 4.3, Refitting the camshafts

Refit the timing belt pulleys and the timing belts	N 4.2, Refitting the timing belt covers
Refit the lubrication system	N 2.1, Refitting the oil pump
Refit the engine to the frame	N 1, Refitting the engine